TOPICS IN COGNITIVE PSYCHOLOGY: PSYCHOLOGICAL INTERVENTIONS USING DIGITAL TECHNOLOGIES
PSY.875

SMARTPHONE APPS FOR DEPRESSION AND ANXIETY: A SYSTEMATIC REVIEW AND META-ANALYSIS OF TECHNIQUES TO INCREASE ENGAGEMENT

Review article

ALMA AGUILAR RAMÍREZ
JOSÉ MANUEL PÉREZ TEXEIRA

INDEX

1 ***INTRODUCTION***
- BACKGROUND AND RESEARCH PROBLEM
- GAP AND OBJECTIVES

2 ***FRAMEWORKS***
- PSD
- BEHAVIORAL ECONOMICS

3 ***METHODS***
- STUDY DESIGN AND SELECTION

4 ***RESULTS***
- EFFICACY
- ENGAGEMENT

5 ***DISCUSSION AND CONCLUSION***

INTRODUCTION

BACKGROUND AND RESEARCH PROBLEM

DIDITAL MENTAL HEALTH APPS

- Rapid growth of apps for depression and anxieaty.
- +10.000 apps currently available
- Increased accesibility and lower cost

PREVIOUS EVIDENCE

Meta-analyses: affective in reducing sympotms

MAIN PROBLEM

- Very low real-world adherence
- Only 3.9 % rentention after 15 days

THE CHALLENGE ES NOT EFFECTIVENESS, BUT SUSTAINED ENGAGEMENT

INTRODUCTION

RESEARCH GAP & OBJECTIVES

RESEARCH GAP

- What makes apps engaging?
- Limited focus on design features influencing adherence and outcomes

STUDY OBJECTIVES

1. Systematic review of standalone apps for depression and anxiety
2. Identify engagement strategies (PSD, Behavioral Economics)
3. Examine link between strategies, engagement and clinical outcomes

FRAMEWORK

PERSUASIVE SYSTEM DESIGN (PSD)

technological systems specifically designed to influence user behavior and maintain engagement. PSD is particularly relevant in digital mental health because smartphone interventions often lack direct human support from therapists or clinicians.

Primary Task Support

- self monitoring
- personalization and tailoring

Dialogue Support

- reminders
- feedback
- encouragement
- rewards

Social Support

- comparison
- cooperation
- competition

Credibility Support

- trust
- expertise
- reliability

FRAMEWORK

BEHAVIORAL ECONOMICS AND GAMIFICATION

Behavioral economics	focused on understanding ways that human decision-making differs from what would be expected in traditional economic models of “rational” actors	avoiding losses, “starting fresh,” committing to action, and by lotteries
Gamification	the use of game-like experiences in non-game services, to further improve user engagement	points, badges, levels, and avatars

These strategies were rarely systematically evaluated in mental health

METHODS

STUDY DESIGN

- Systematic review and meta-analysis
- PRISMA guidelines
- Protocol registered in PROSPERO

Datas sources: Medline, Embase, PsycINFO, PsycArticles, CINAHL, Cochrane Library, AMED

Analysis approach

Random-effects model used
Accounts for variability across studies (designs, populations, interventions)

CHARACTERISTICS OF INCLUDED STUDIES

- 4159 participants
- randomized controlled trials
- Mean age: 32.9 years
- Majority female
- Intervention duration: 2-12 weeks

Engagement measures

Engagement is a multidimensional construct and makes cross-study comparison difficult

METHODS

INCLUSION CRITERIA AND STUDY SELECTION

- Standalone smartphone apps for depression/anxiety
- Participants with symptoms (diagnosis or validated scales)
- At least one engagement measure

6287 records identified // 25 included

Table 1. Study characteristics.

Type of study ($N = 25$)	
RCT	15 articles
Pilot/quasi experimental	10 articles
Gray literature ($N = 3$)	
Dissertation	2 articles
Pre-Print	1 article
Apps ($N = 29$)	
Participant demographics ($N = 4159$)	
Average age (Min–Max) (years)	32.9 (13–76)
Female (%)	66.6
Diagnosis of study participants	
Major depressive disorder	19 studies
Generalized anxiety disorder	5 studies
Social anxiety disorder	3 studies
Dysthymia	1 study
Other depressive disorder	1 study
Method of symptom evaluation	
Self-identification	2 studies
Validated self-assessment measure	19 studies
Clinician-administered measure or medical record	8 studies
Study length (Weeks)	
Average Length of Intervention (Min–Max)	5.82 (2–12)

METHODS

QUALITY

- Mixed quality overall
- Quasi-experimental studies: low/moderate quality
- RCTs: some risk of bias

HETEREOGENITY

- High variability ($I^2 > 75\%$)
- Due to differences in apps, samples, and outcomes

PUBLICATION BIAS

Possible overrepresentation of positive results

- Egger's regression
- Trim-and-fill

RESULTS

Characteristics of the Apps

- Most were based on cognitive-behavioral therapy
- Several apps also included tracking functions
- few apps include safety features (e.g., crisis support, suicide risk monitoring)
- lack clear privacy policies or data protection

RESULTS

PSD & Behavioral Economics Features

Table 3. Average persuasive system design features and categories.

Persuasive system design feature	Mean (SD)	Min	Max
Primary task support	3.30 (1.26)	1	6
Dialogue support	1.93 (1.20)	0	4
Social support	0.37 (0.93)	0	4
Credibility support	0.90 (1.24)	0	4
Total features	6.50 (2.60)	2	12
Total categories	2.50 (0.90)	1	4

Total features in each category = 7; total categories = 4.

RESULTS

Overal efficacy

In pre-post analyses, the effect size was moderate g = 0.6217 In randomized controlled trials, the effect was smaller but still statistically significant. g = 0.2888

Clinical Outcomes

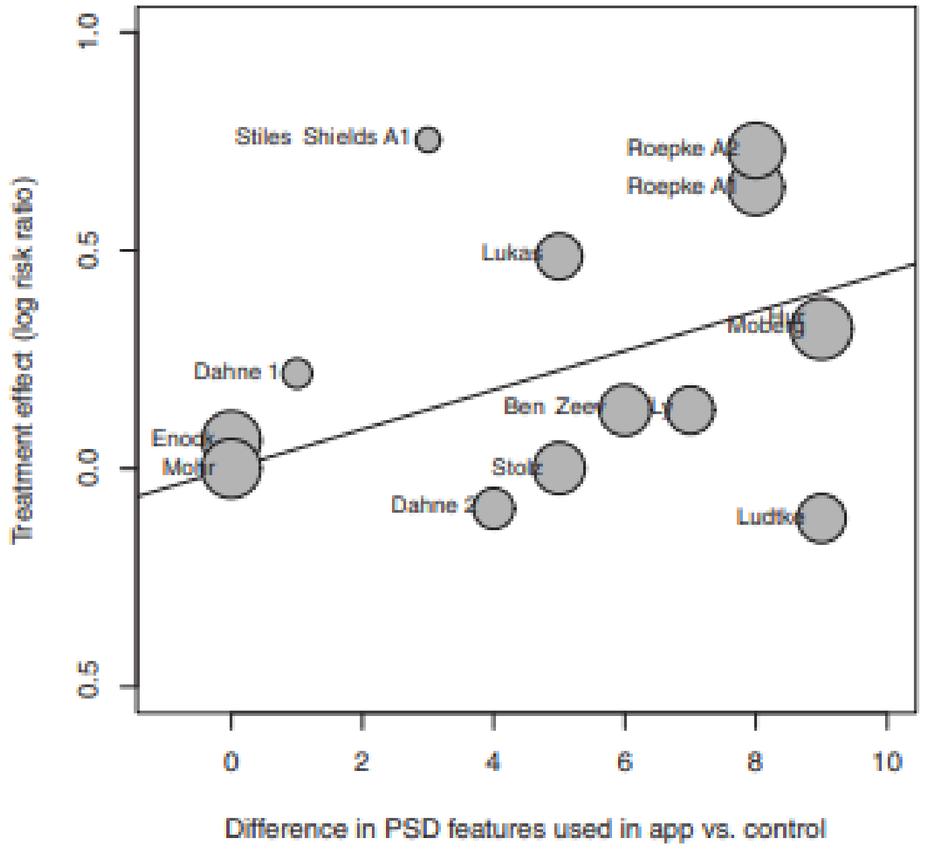

Fig. 4 Mixed effects meta-regression analysis of difference in PSD features and efficacy for app vs control. Bubble plot of mixed effects meta-regression analysis of difference in PSD features and efficacy for app vs control.

RESULTS

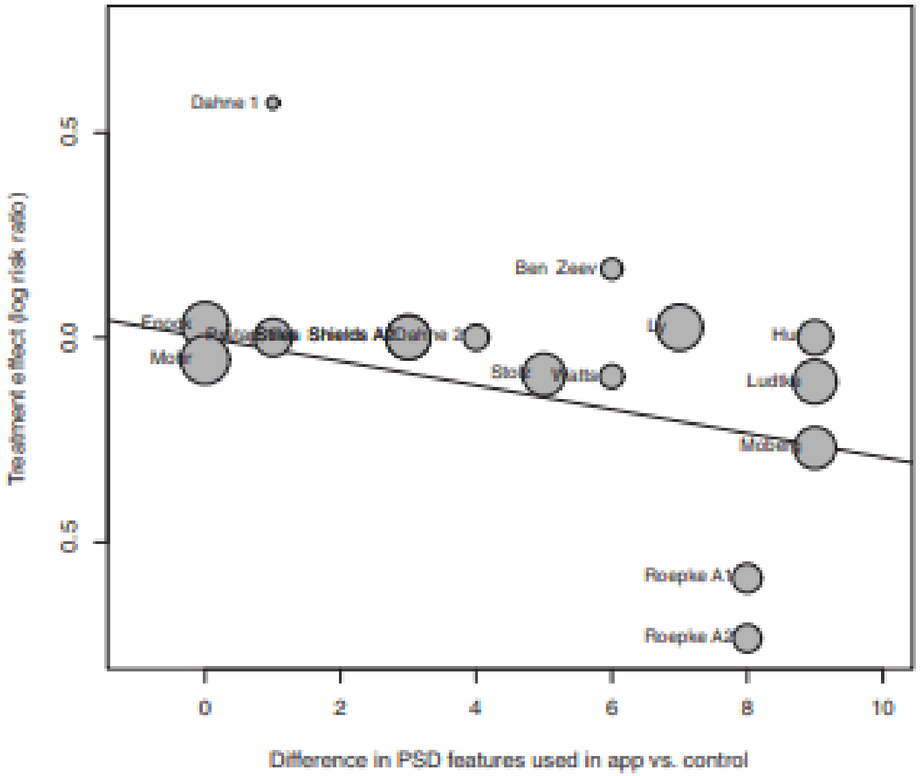

Fig. 5 Mixed effects meta-regression analysis of difference in PSD features and study completion rate for app vs control. Bubble plot of mixed effects meta-regression analysis of difference in PSD features and study completion rate for app vs control.

Completion Rates

Several possible explanations:

- improvement more rapidly – discontinue the intervention earlier.
- not adequately reflect meaningful engagement.

Exploratory analyses

dont support that greater efficacy explained the lower completion rates

RESULTS

Quality vs Quantity

- Engagement should not only in terms of duration but quality of engagement
- Benefit substantially from an app without completing every module
- Important for future studies should incorporate more sophisticated engagement

STAKEHOLDERS

PATIENTS

- Better access to mental healthcare
- More engaging interventions

CLINICIANS

- Complement traditional therapy
- Monitor symptoms and progress

DEVELOPERS AND UX DESIGNERS

- Apply PSD principles
- Improve engagement and usability

RESEARCHERS

- Improve engagement measurement
- Guide future studies

HEALTHCARE SYSTEMS & POLICYMAKERS

- Scalable and cost-effective interventions
- Increase access to care

DISCUSSION

LIMITATIONS AND CONCLUSION

- Mental health apps are effective, although the effects are modest.
- Persuasive design features appear to improve clinical outcomes
- Social support features and behavioral economics strategies are still underused

DISCUSSION

DISCUSSION

LIMITATIONS AND CONCLUSION

LIMITATION

- *High heterogeneity* across studies, different in design, duration, and populations
- The methodological quality was variable, showing some concerns or high risk of bias
- *Engagement* was measured in very *inconsistent* ways
- Publication bias suggests that *positive results* may be *overrepresented*
- Many relied on self-report measures instead of clinical interviews

DISCUSSION

LIMITATIONS AND CONCLUSION

CONCLUSIONS

- Smartphone apps can reduce symptoms of depression and anxiety modestly
- the first meta-analysis examining the relationship between engagement strategies and clinical efficacy.
- Detailed understanding of app engagement.
- Effectiveness depends not only on therapeutic content, but also on sustained user engagement

THANK *You!*